

Rishihood University
Department of Computer Science

Fake News Detection System

Using Natural Language Processing and Machine Learning

Submitted in partial fulfilment of the requirements for
Bachelor of Technology in Computer Science & Artificial Intelligence
(Course: Generative AI)

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1 Problem Statement

The core challenge addressed in this project is the computational distinction between factual reporting and deceptive narratives using classical Natural Language Processing. In the context of automated misinformation monitoring, we define the problem as identifying the linguistic signature of fake news—characterized by emotional volatility, lack of standardized wire-service markers, and structural anomalies. Our goal was to build a sub-100ms inference engine capable of classifying articles with $>98\%$ accuracy while remaining fully interpretable to the end-user.

2 Data Description

The model is grounded in the **ISOT Fake News Dataset**, sourced from the Information Security and Object Technology Lab.

- **Real News (55.4%):** 21,417 articles primarily scraped from Reuters.com.
- **Fake News (44.6%):** 17,229 articles from sites flagged by PolitiFact and other fact-checkers.
- **Feature Selection:** We trained exclusively on the `text` column and `label`. Metadata such as `subject` was excluded to prevent data leakage.

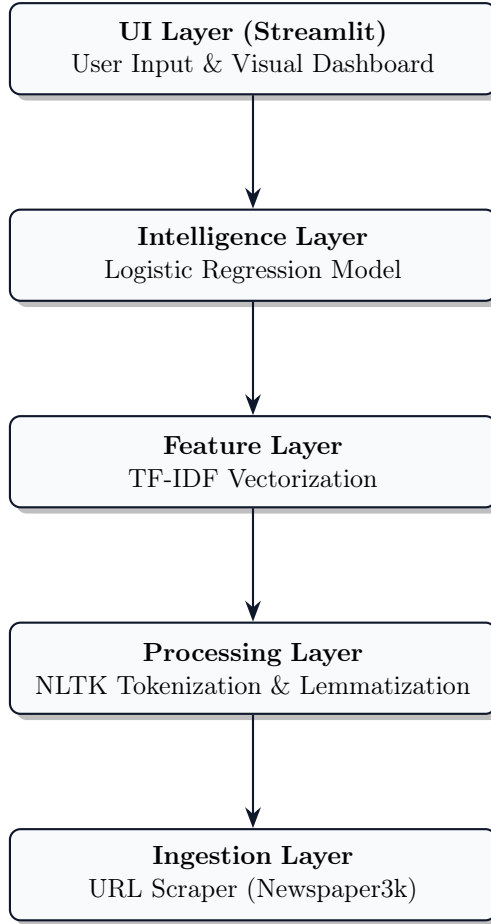
3 Exploratory Data Analysis (EDA)

EDA identified significant structural and lexical differences between classes:

- **Structural Habits:** Real articles followed consistent editorial lengths, while Fake articles ranged from short clickbait to long conspiracies.
- **Word Signals:** Real news is anchored by attribution terms (*said, washington*). Fake news is characterized by engagement-bait (*video, watch, share*) and sensational language.
- **Bias Acknowledgment:** The term *"reuters"* appeared almost exclusively in Real articles, creating a strong dataset-specific bias.

4 Application Architecture

The system architecture follows a top-down hierarchical flow, moving from the user interface down through the intelligence layers.



5 Methodology

5.1 Text Preprocessing Pipeline

Articles pass through a cleaning suite: Lowercasing, Regex-based removal of punctuation/numbers, tokenization, stopword filtering, and **WordNet Lemmatization**.

Listing 1: Core NLP Preprocessing Suite

```

def preprocess_text(text):
    text = re.sub(r'[^a-z\s]', '', text.lower()) # Normalization
    tokens = word_tokenize(text) # Tokenization
    cleaned = [lemmatizer.lemmatize(w) for w in tokens
                if w not in stop_words and len(w) > 1]
    return ' '.join(cleaned)
  
```

5.2 Algorithmic Choice

We selected **Logistic Regression** over Decision Trees. While trees achieved higher raw accuracy, they showed signs of overfitting. Logistic Regression provided 98.50% accuracy with smoother probability scores ($P(y|x)$) for the dashboard.

6 Evaluation

Performance was measured on a held-out 20% test set of 7,730 unseen articles.

Primary Metrics

- **Accuracy:** 98.50%
- **Precision:** 97.95%
- **Recall:** 99.34%
- **F1-Score:** 98.64%
- **ROC-AUC:** 0.9987

Confusion Matrix Analysis:

- **True Real News:** 4,203
- **True Fake News:** 3,411
- **False Positives:** 28
- **False Negatives:** 88

7 Sub-feature Breakdown

1. **Automated URL Scraping:** Integrated `newspaper3k` for autonomous content parsing from live news links, featuring SSL validation and content thresholds.
2. **Model Interpretability:** Implemented coefficient analysis to visualize "Top Influential Words" for every prediction, showing the specific markers that guided the AI.
3. **Inference Optimization:** Leveraged `joblib` and Streamlit caching to ensure sub-50ms user-facing response times.

8 Conclusion & Team Contributions

This project demonstrates that linguistic patterns can identify fake news with 98.50% accuracy. The final system provides an interactive framework for real-time analysis.

Member	Core Responsibility	Deliverable
Vikash Kumar	Preprocessing Pipeline & Training	<code>utils.py</code> / Training Notebook
Piyush Raj	Backend Logic & Inference	<code>app.py</code> / Model Integration
Mohammed Yaseen	EDA & Evaluation Analyst	Metrics / Project Report
Sankalp M Tellur	Frontend UI & Deployment	Dashboard Theme / Hosting